Table 8.1 Evolution of drone roles in modern warfare.

Generation/ stage	Primary role	Key capability	Technological enabler	Representa UAV syste
Early UAVs (pre-2000)	Basic reconnaissance and battlefield observation	Limited remote camera imaging and short-duration surveillance	Radio-controlled flight systems and basic telemetry	MQ-1 Pred IAI Searche
Networked UAVs (2000–2015)	Intelligence, surveillance, reconnaissance (ISR) with precision strike capability	Persistent surveillance and guided weapon delivery	Satellite communication, EO/IR imaging systems, laser designation	MQ-9 Reap IAI Heron, Harpy
Semi-autonomous UAVs (2015–2022)	Integrated combat support and target coordination	Automated target tracking, patrol route optimization, and partial onboard AI	Embedded microprocessors, machine-vision sensors, advanced GNSS navigation	Bayraktar T CH-4, Wing Loong II
Fully autonomous and swarm-enabled UAVs (2022–Future)	Cooperative combat operations, swarm-based offensive maneuvers	Self-organizing multi-drone decision systems and dynamic battlefield adaptation	Edge AI computing, swarm algorithms, secure datalink coordination	Loyal Wingman (Boeing), Sukhoi S-7(Okhotnik, ALTIUS, Shahed-13(